

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (AI & Machine Learning) SEMESTER - 7 Winter 2025 (Regular)

Course :Bachelor of Technology (Artificial Intelligence & Machine Learning) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTAIC701 - NATURAL LANGUAGE PROCESSING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which of the following is the main purpose of tokenization in NLP?A) a) Remove rare wordsB) Convert text into individual units like words or sentencesC) Convert text to lowercaseD) Remove punctuation
 - 2 Minimum edit distance is used to measure:A) Frequency of words in a documentB)) Semantic similarityC) The number of operations required to transform one string into anotherD) Probability of word sequences
 - 3 TF-IDF gives high weight to words that are:A)) Very frequent in all documentsB) Rare across documents but frequent in a specific documentC) Stop wordsD) Grammatically correct
 - 4 Word2Vec is mainly used for:A)) Syntax parsingB) Converting words into dense numerical vectorsC) Classifying documentsD) Removing stop words
 - 5 Transformers primarily rely on which mechanism.A) Convolution B) Recurrent connections C) Attention mechanism D) Minimum edit distance
 - 6 Encoder-Decoder architecture is widely used in:A) Stop-word remove l B) Translation and text summarization C) Tokenization D) Regularization
 - 7 LSTM networks are mainly designed to solve which problem?A) OverfittingB) OverfittingC) Tokenization errorsD) Sentence segmentation
 - 8 In MEMM which major limitation arises due to conditioning on the previous predicted label?A) Exposure bias B)Label bias problemC) Vanishing gradient problem D) Data sparsity problem
 - 9 Constituency parsing using CKY algorithm requires the grammar to be converted into.A) Dependency Normal Form B) Head-Driven Phrase Structure GrammarC) Chomsky Normal Form D) Gribiche Normal Form
 - 10 Logistic regression is mainly used for.A) Clustering B) Classification C) Translation D) Speech recognition
 - 11 Word2Vec Skip-gram model, which factor MOST strongly influences the quality of word vectors?A) The number of POS tags used during trainingB) The window size that defines the context around each target wordC) The number of stop words removed before trainingD) The size of the dependency tree generated for each sentence

- 12 In LSTM networks, which component is primarily responsible for preventing the vanishing gradient problem by preserving information across long sequences? A) a) Input gate B) Forget gate C)) Output gate D) Cell state (memory cell)
- Q2. Solve the following.
- A) What do mean by Minimum Edit Distance in NLP. explain with example 6
- B) Explain Tokenization and segmentation in detail. 6
- Q3. Solve the following.
- A) Explain TF-IDF in detail with application 6
- B) Explain Smoothing Techniques in detail 6
- Q4. Solve Any Two of the following.
- A) Discuss two commonly used metrics for evaluating the performance of NER systems. 6
- B) Define parsing ,explain Dependency Parsing. Describe its types, and advantages with a suitable example. 6
- C) Given three documents: oD1: 'I love natural language processing' oD2: 'Language models are powerful' oD3: 'I love machine learning and models' Construct the TF table, compute IDF, and calculate the TF-IDF weight of the term “models” in each document. 6
- Q5. Solve Any Two of the following.
- A) In a corpus of 10,000 words, the word “deep” occurs 100 times, “learning” occurs 80 times, and the bigram “deep learning” occurs 50 times. Calculate PMI(deep, learning) and interpret the result 6
- B) What is Text Classification? Explain How Does Text Classification Work? 6
- C) Define multinomial logistic regression and its application in text classification 6
- Q6. Solve Any Two of the following.
- A) Define Simple Recurrent Networks and explain their has it architecture. What is the main imitation of SRNs in handling long-term dependencies? 6
- B) Explain text summarization and multiple document text summarization with neat diagram 6
- C) Describe the encoder-decoder model using RNNs. Provide an example of a sequence-to-sequence task where the encoder-decoder architecture is beneficial. 6

***** End *****